



Azure AI Services Infrastructure

Enterprise-Ready AI Platform with Network Isolation & Private Endpoints



Virtual Network: ai-platform-vnet-region (192.168.0.0/16)



agent-subnet

192.168.0.0/24

Delegated to Microsoft.App/environments

- ★ AI Agent Runtime Environment
- 🔗 Network Injection Target
- 🏭 Capability Host Deployment



pe-subnet

192.168.1.0/24

Private Endpoints Subnet

Network isolation for all services

DNS resolution hub



Private Endpoints (in pe-subnet)



AI Services
192.168.1.8-10



Search Service
192.168.1.6



Cosmos DB
192.168.1.4-5



Storage Account
192.168.1.7



Private DNS Zones

- privatelink.cognitiveservices.azure.com
- privatelink.openai.azure.com
- privatelink.services.ai.azure.com
- privatelink.search.windows.net
- privatelink.documents.azure.com
- privatelink.blob.core.windows.net





Azure AI Services Hub

ai-services-platform



AI Foundry Hub Service:



Project: ai-platform-project



Model: GPT-4o (GlobalStandard)



Capability Host: Agents

Connected Resources:

- Vector Store: ai-search-platform
- Storage: ai-storage-platform
- State Store: ai-cosmos-platform
- Network Injection: agent-subnet



External Connections:

- Microsoft Fabric Workspace
- Databricks Genie Analytics
- Power BI Premium Datasets
- Data Lake Gen2 Sources



Security Features:

- Content Safety (DefaultV2 Policy) • AAD Authentication • Private Endpoint Only • System Assigned Identity



Azure AI Search

ai-search-platform

Configuration:

- Standard SKU (1 replica, 1 partition) • Vector Store for RAG Operations • System Assigned Identity • AAD Authentication Enabled • Public Network Access: Disabled • Semantic Search: Disabled



Network Security:

- Private Endpoint: 192.168.1.6 • DNS Zone: privatelink.search.windows.net • Connection State: Approved



Azure Cosmos DB

ai-cosmos-platform

State Store Configuration:

- SQL API with Session Consistency • Database: platform_memory • Autoscale: 100-1000 RU/s per container • TLS 1.2+ Only, Local Auth Disabled



State Store Containers:

- agent-entity-store (Agent persistence) • conversation-state-store (Conversation state) • system-state-store (System state)



Security:

- Role-based Access Control • Private Endpoint: 192.168.1.4-5 • Geo-redundant Backup



Azure Storage Account

ai-storage-platform

Storage Configuration:

- StorageV2, Standard ZRS • Services: Blob, File, Queue, Table • Hot Access Tier • TLS 1.2+ Only



Containers:

- platform-blobstore (Project artifacts) • platform-agent (Agent runtime data)



Security Features:

- No Public Blob Access • Shared Key Access: Disabled • Private Endpoint: 192.168.1.7 • Cross-tenant Replication: Disabled

- AI/ML Services
- Search & Vector Store
- Database Services
- Storage Services
- Agent Runtime Subnet
- Private Endpoints Subnet
- Private Endpoints